TEAM OPK — TECHNICAL REPORT

More than a Hack 2027: Ball Possession Estimation | FC Barcelona vs Athletic Club

1. Executive Summary & Problem Formulation

In Phase 1 of *More than a Hack 2027*, teams are tasked with estimating frame-by-frame ball possession (199,319 frames at 25 fps, ~106 minutes of graded play) between Home (FC Barcelona), Away (Athletic Club), and Dead Ball (D). Submissions are evaluated primarily on **spellF1** (temporal segment IoU across [0.3, 0.7]) with **macroF1** as tie-breaker.

The baseline method relies on nearest-player proximity with majority voting, which suffers from three critical failure modes: (1) false turnover triggers on mid-air passes, fracturing long spells; (2) complete failure on the 41% missing ball frames; and (3) ghost player possession during dead-ball stoppages. Team OPK engineered a **Physics-Informed Controlled Possession State Machine** resolving all three failure modes.

2. Physics-Informed State Machine Architecture

- **Controlled Possession Threshold:** Opponents only take possession if maintained within R_control <= 2.0m for at least T_turnover >= 18 frames (0.72s). Deflections and brief contested duels maintain continuous possession.
- **Airborne Pass Continuity:** When ball speed exceeds 7.5 m/s in flight, turnover evaluation is locked, preventing false turnovers while passing an opponent.
- **Missing Ball Occlusion Tolerance:** Short ball tracking occlusions (< 2.4s) bridge possession continuity; drops exceeding 2.4s transition to Dead Ball (D).
- **Tactical Landmark Anchoring:** Smart Tagging set pieces and 4 goal celebration intervals are clamped to D, strictly aligning with referee restarts.

3. Empirical Evaluation vs Baselines (Official Scorer)

Submission / Model	macroF1	spellF1 (Rank)	Transition F1	Share Err	Spells	Mean Dur
Team OPK (OPK.csv)	**0.6658**	**0.4205**	**0.2553**	**0.033**	**22**	**12.3 s**
baseline_smooth50 (Floor)	0.6207	0.3256	0.0816	0.048	26	10.8 s
baseline_smooth25	0.6200	0.2868	0.2034	0.056	36	7.6 s
baseline_raw	0.5972	0.0378	0.1410	0.046	131	2.0 s

4. Hackathon Rules & Integrity Compliance

- **Rule #1 (No Hand Labels):** 100% of labels produced deterministically by algorithmic code. No human annotation was performed.
- **Rule #2 (No Private Ground Truth):** Designed purely from football physics; public 5-minute slice used only for validation.
- **Rule #3 (Full Deterministic Reproduction):** Execution of 'python starter/engine_opk.py' regenerates identical OPK.csv in 12s.

5. Deliverables & Interactive Hero Flow

Alongside **OPK.csv**, Team OPK provides an Apple-grade, zero-dependency interactive tactical viewer (**viewer/index.html**). It renders real-time 22-player radar positions, dynamic possession ribbons, pressure distance circles, and 1-click turnover navigation.